UPetu: A Unified Parameter-Efficient Fine-Tuning Framework for Remote Sensing Foundation Model

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***Abstract*— Recent advancements in remote sensing foundation models have unveiled their tremendous potential in addressing Earth observation tasks. Presently, when large-scale foundation models are transferred to downstream tasks, the prevalent approach is to adopt the full-tuning strategy, resulting in significant increases in storage demands and computational costs. Although the introduction of parameter-efficient fine-tuning (PEFT) has mitigated this issue to some extent, mainstream PEFT methods are primarily designed for classification tasks and often prove insufficient to meet the demands of dense prediction tasks. To overcome the aforementioned limitations, we propose a unified PEFT framework UPetu, encompassing two essential and complementary modules: the efficient quantization adapter module (EQAM) and the context-aware prompt module (CAPM). EQAM is specifically designed to enhance the correlation between fine-grained feature information and task-specific knowledge through the introduction of quantization linear (Q-Linear) layers and nonlinear activation functions. In addition, CAPM is introduced to acquire rich contextual features by incorporating trainable prompts into multiscale features. The synergistic integration of both the modules enhances the representation learning capability and generalization transferability of the foundation model. Extensive experiments on three remote sensing scene classification datasets demonstrate the superiority of UPetu over other fine-tuning methods. With the update of only 0.73% of ConvNeXt-B parameters, our UPetu achieves superior performance compared with full-tuning on the UCM-55, AID-28, and AID-55 datasets. Furthermore, experiments conducted on semantic segmentation and change detection tasks provide additional evidence of the effectiveness and generalization capabilities of the proposed UPetu.**

***Index Terms*— Foundation model, parameter-efficient fine-tuning (PEFT), remote sensing.**

I. Introduction

With the rapid development of Earth observation [1] and deep learning technologies, large-scale vision models gradually dominate the field of remote sensing over the past few years [2], [3], [4]. Following the principles of transfer learning, downstream tasks can substantially profit from the knowledge learned via pretrained models. Early pretrained models such as CNNs [5] and transformers [6] can be pretrained via supervised or self-supervised learning. Recently, toward unifying visual understanding tasks, advanced pretraining paradigms such as MoCo [7] and MAE [8] have been proposed, which are known as foundation models [9] as a whole.

The foundation models typically adhere to the "Pretrain & Fine-tune" paradigm, as shown in Fig. 1, wherein the foundation models were initially pretrained on extensive data through self-supervised learning to learn universal representations. Subsequently, fine-tuning is used to transfer the pretrained model's understanding capabilities, thus achieving promising performance on downstream tasks.

As the prevailing adaptation strategy, full-tuning necessitates the maintenance of distinct parameter sets for each task and deployment scenario, which equate in size to the pretrained foundation models. These parameters operate independently and cannot be shared, rendering them resource-intensive for real-world applications. Furthermore, it becomes exceedingly storage-expensive, as storage costs increase linearly with the expanding array of downstream tasks and evolving deployment environments. It is particularly salient when deploying large-scale vision foundation models, such as ViT-Huge [10] (632 M parameters), on mobile systems. In contrast, while only fine-tuning the downstream task heads avoids updating the entire backbone model, it often results in unsatisfactory performance.

Notably, taking inspiration from natural language processing (NLP) [11], [12], an emerging solution for pretrained vision models is to replace full-tuning with parameter-efficient fine-tuning (PEFT), which only tunes a small number of trainable parameters while freezing the majority that is shared by multiple downstream tasks. Furthermore, by fine-tuning a more limited set of parameters, it not only alleviates optimization complexities but also effectively combats overfitting when adapting large-scale foundation models to specific target datasets, ultimately resulting in performance that matches or even surpasses that of full-tuning. However, the mainstream visual PEFT methods are primarily designed for image classification tasks, neglecting the multiscale characteristics inherent in dense prediction tasks such as semantic segmentation and change detection. Furthermore, the challenge of considering the prior information in remote sensing data samples and spatio-temporal correlations in multiscale features during the fine-tuning process remains significant.

To address the aforementioned shortcomings, we propose a **u**nified **p**arameter-**e**fficient fine-**tu**ning framework **UPetu**, which is the first work to explore PEFT solutions of remote

Manuscript received 25 January 2024; revised 8 March 2024; accepted 21 March 2024. Date of publication 29 March 2024; date of current version 5 April 2024. This work was supported by the National Key Research and Development Program of China under Grant 2022ZD0118401. *(Corresponding author: Tianzhu Liu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3382734

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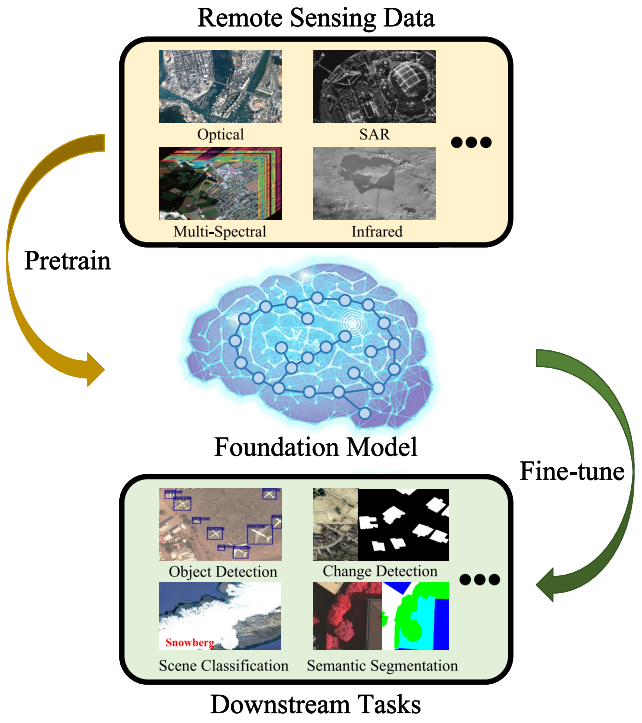


Fig. 1. Illustration of the "Pretrain & Fine-tune" paradigm for remote sensing foundation model. Foundation models pretrained on multimodal remote sensing data can benefit various downstream tasks. Meanwhile, fine-tuning uses the knowledge acquired from the source domain to facilitate the learning process of the target domain.

sensing foundation models across various downstream tasks, encompassing scene classification and dense predictions. When fine-tuning the remote sensing foundation model on downstream tasks, the parameters of the foundation model remain frozen, with only the parameters of UPetu being tunable. Specifically, our proposed UPetu framework primarily comprises two complementary modules: the efficient quantization adapter module (EQAM) and the context-aware prompt module (CAPM). EQAM primarily incorporates two quantization linear (Q-Linear) layers and nonlinear activation, facilitating the interaction between fine-grained feature information and task-specific knowledge. Meanwhile, CAPM introduces a set of trainable parameters enriched with comprehensive contextual information into the foundation model through engagement with multiscale features.

Overall, the main contributions of this work are summarized as follows.

1) We propose a novel UPetu framework to reduce the computational and storage costs incurred when fine-tuning the remote sensing foundation model on downstream tasks. To the best of our knowledge, this is the pioneering endeavor to introduce the PEFT method in remote sensing field, effectively addressing both scene classification and dense prediction tasks simultaneously.
2) An EQAM is proposed to learn task-specific feature representations during the fine-tuning process. Moreover, a computationally efficient quantization method is introduced in the feedforward linear layers to strike a balance between efficiency and accuracy.
3) A CAPM is designed to dynamically generate trainable conditionally input-dependent prompts and interact with feature representations to enable the foundation model to better comprehend and adapt to diverse remote sensing tasks.

The subsequent sections of this article are organized as follows: an exploration of related work on remote sensing foundation models, prompt tuning methods, and adapter tuning methods is presented in Section II. Our proposed UPetu framework, along with detailed explanations of EQAM and CAPM, is introduced in Section III. Section IV showcases the experimental settings and provides extensive results. Finally, Section VI encapsulates the concluding remarks of this article.

II. Related Work

A. Remote Sensing Foundation Model

Remote sensing foundation models serve the critical purpose of providing scalable and versatile approaches for handling diverse and complex remote sensing data. By leveraging self-supervised learning methodologies on extensive and unlabeled remote sensing data, foundation models adeptly extract intrinsic patterns, features, and representations inherent in remote sensing images, enabling their application across a diverse range of downstream tasks. Prominent foundation models in remote sensing are primarily constructed based on two key techniques: contrastive learning and generative learning.

Contrastive learning aims to augment the similarity between feature representations derived from two different views of the same image. DINO-MC [13] used multiple views with varying sizes for self-supervised learning within the DINO framework [14]. GASSL [15] incorporated geo-location prediction as an auxiliary pretext task within the MoCo-v2 framework [16]. A novel contrastive learning method was proposed by SeCo [17] to perceive short-term and long-term changes through the utilization of the spatiotemporal structure present in temporal remote sensing image sequences. Besides, SSCCL [18] additionally introduced a consistency branch within the contrastive learning framework to model spatial and semantic correlations. SkySense [19] innovated by introducing a billion-scale multimodal remote sensing foundation model with a factorized multimodal spatiotemporal encoder, multigranularity contrastive learning for diverse modalities, and geo-context prototype learning for region-aware features.

Generative learning primarily aims to predict and restore masked image pixels through masked image modeling (MIM), facilitating coherent reconstruction of the image. RingMo [2] modified the masking strategy in MAE [8] to accommodate dense objects in remote sensing images. RingMo-Sense [20] introduced triple-branch prediction networks with parameter-sharing and progressive joint training, innovatively tackling the spatial and temporal multiscale challenges in remote sensing data for spatiotemporal prediction. SatMAE [21] introduced innovative masking and temporal embedding techniques for unsupervised pretraining on satellite imagery. Besides, RVSA [3] innovatively presented rotated varied-size window attention to large vision models, enhancing computational efficiency and object representation in remote sensing tasks.

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B. Prompt Tuning

Prompt tuning [22], [23] is introduced for the first time in the domain of NLP, whereby downstream datasets are reformulated as masked language modeling tasks, enabling the frozen foundation model to directly adapt to new tasks without requiring any parameter updates. Taking inspiration from prompt techniques commonly used in NLP, prompt tuning is now being introduced into the domain of computer vision (CV). The goal of prompt tuning is to modify the input distribution, thereby enhancing the relevance of the feature representations learned by the foundation model during the pretraining period to downstream tasks. Formally, the mathematical formulation can be expressed as follows:

$$\mathcal{P} = [\mathcal{T}, a_1, \ldots, a_i, \ldots, a_n] \tag{1}$$

where $\mathcal{P}$ denotes the vision-driven prompts, $\mathcal{T}$ denotes the embeddings of tokens outputted by the frozen foundation model, and a_i is the ith learnable vector.

The existing extensive studies [24], [25], [26], [27], [28] aim to reduce the disparity between pretraining and downstream tasks by altering the distribution of input images. The VPT [24] method inserted a set of learnable parameters, treating them as an integral part of input tokens, thereby guiding the pretrained foundation model to perform various downstream tasks. Similarly, DePT [25] incorporated learnable visual prompts into the vision transformer, focusing the optimization solely on these source-initialized prompts, all the while maintaining the frozen state of the vision transformer. In addition, ZegCLIP [26] introduced learnable vectors as visual prompts at every layer of the frozen CLIP [27] image encoder, enabling the foundation model to handle zero-shot semantic segmentation task. Moreover, LPT [28] fine-tuned shared prompts for uncovering general features within the comprehensive long-tailed dataset, while also optimizing group-specific prompts to enhance the frozen, pretrained vision models' ability for fine-grained discrimination. Although the aforementioned studies used a straightforward approach to formulate visual prompts, significant advancements have been achieved in transferring discerning discrimination and enhancing generalization.

To further explore the effectiveness of visual prompts, some works [29], [30], [31], [32], [33], [34] attempted to design a subnetwork for constructing visual prompts. NOAH [29] used a neural architecture search algorithm to automatically determine the feature dimension for the adapter and the token length for the vision prompt. FPTrans [30] used a pretrained ViT network with an added lightweight module to segment the input image into foreground and background regions, enabling the attachment of distinct foreground and background prompts. Fine-grained retrieval prompt tuning (FRPT) [31] constructed a lightweight sampling network to selectively focus on discriminative regions within input images, resulting in the generation of specific vision prompts. Objects within video frames were localized as vision prompts by RePro [32], using a tracklet detector. Subsequently, the correlation between subjects and objects was learned based on the acquired vision prompts. Gan et al. [33] acquired domain-specific prompts and domain-agnostic prompts, thereby capturing the knowledge from the source domain while preserving the domain-shared knowledge during continual adaptation. Multiple regions of interest, regarded as vision prompts, were generated by ViLD [34] through the region proposal network to facilitate the alignment of their visual embeddings and textual embeddings for open-vocabulary object detection. These efforts generate suitable prompts for downstream tasks and enhance the interpretability of these visual prompts.

C. Adapter Tuning

Adapter tuning aims to introduce additional trainable parameters into a frozen pretrained model to facilitate learning for downstream tasks. Adapters were initially proposed by Houlsby et al. [35] for NLP tasks and have gradually been applied to the CV domain.

Typically, adapters consist of a linear down-projection, a nonlinear activation function, an up-projection, and a residual connection. For a given input feature map $\mathbf{F}^l$, the adapter module initially uses a down-projection using $\mathbf{W}_{\text{down}}$ to project the feature onto a lower dimensional representation. Subsequently, a nonlinear activation function is applied, followed by an up-projection using $\mathbf{W}_{\text{up}}$. The formulation is articulated as follows:

$$\hat{\mathbf{F}}^l = \delta\left(\text{Norm}\left(\mathbf{F}^l\right)\mathbf{W}_{\text{down}}\right)\mathbf{W}_{\text{up}} \tag{2}$$

where $\mathbf{W}_{\text{down}}$ and $\mathbf{W}_{\text{up}}$ denote two learnable linear projection matrices, δ represents the nonlinear activation function, Norm is the normalization layer, and $\hat{\mathbf{F}}^l$ signifies the optimized features outputted by the adapter module.

Recent studies have attempted to insert lightweight adapter modules into pretrained vision foundation models and fine-tune them while keeping the pretrained weights frozen. Res-adapt [36] developed an adapter residual module to dynamically adjust the large-scale model to different visual tasks in real-time. Polyhistor [37] divided a hypernetwork into two distinct hypernetworks and transformed an adapter weight matrix into two kernels to reduce parameters in a multitasking architecture. Semantic information from diverse levels was harnessed by Pro-tuning [38] to enhance the feature space through the creation of multiple stagewise prompt blocks. These blocks can be easily incorporated into a foundation model to enable rapid adaptation to a new task. Besides, Shysheya et al. [39] introduced an innovative method named Fit, which involved scaling and shifting activations and used a Naive Bayes classifier in the final layer for image classification. Moreover, RepAdapter was proposed by Luo et al. [40], introducing a technique that uses reparameterization of sparse structures to approximate neighboring projection weights, further reducing the parameters that need to be fine-tuned for the pretraining model.

III. Proposed Method

We introduce the UPetu framework designed for adapting a pretrained foundation model to downstream tasks in practical remote sensing scenarios. This framework comprises two synergistic modules: the EQAM and the CAPM. During fine-tuning on downstream tasks, UPetu maintains the backbone frozen, while only allowing tunability in EQAMs,

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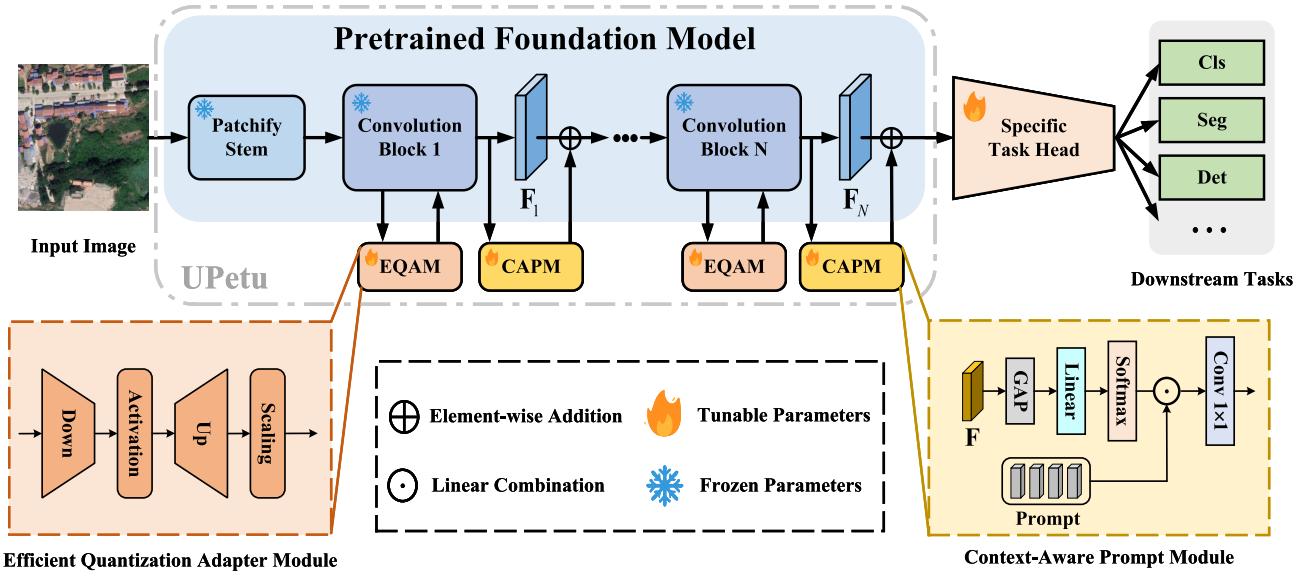


Fig. 2. Overall architecture of our proposed UPetu framework. It mainly consists of three parts: a pretrained backbone, the EQAMs, and the CAPMs. During the fine-tuning stage, we freeze the parameters of the backbone network and only update the parameters of EQAM and CAPM.

CAPMs, and a specific task head. The schematic overview of the proposed UPetu framework is depicted in Fig. 2. In the subsequent sections, we delve into a comprehensive examination of ConvNeXt in Section III-A. Subsequently, we provide detailed insights into the EQAM and CAPM modules in Sections III-B and III-C, respectively.

A. Preliminaries

On a global scale, ConvNeXt seamlessly incorporates a series of architectural concepts derived from vision transformers, all the while upholding the network's intrinsic simplicity akin to a standard ConvNet, devoid of any incorporation of attention-based modules. These discerning design decisions can be succinctly encapsulated as follows:

1) Macro Design: With reference to the design of Swin Transformers, the ratios of computation stages of ConvNeXt are adjusted. This involves the initial adjustment of the computational distribution across stages by altering the specified number of blocks assigned to each stage.

2) Patchify Stem: Motivated by the patchify layer of hierarchical vision transformers, the conventional ResNet-style stem cell is substituted with a patchify layer. This layer is constructed using 4×4 nonoverlapping convolutional layers, complemented by an additional normalization layer designed to enhance training stability.

3) Depthwise Convolution: Within the ConvNeXt block, a depthwise convolution [41] is used—a variation in grouped convolution found in models such as ResNeXt. Aligning groups with channels in this approach facilitates spatial information mixing on a per-channel basis. Similar to the weighted sum in self-attention, this depthwise convolution strategically combines with 1×1 convolution, establishing a distinct separation between spatial and channelwise information integration. This design principle aligns with the philosophy of vision transformers, emphasizing the independent treatment of spatial and channel dimensions.

4) Inverted Bottleneck: In a manner reminiscent of transformers, ConvNeXt introduces an inverted bottleneck architecture in each ConvNeXt block, wherein the hidden dimension of the residual block is expanded to four times that of the input dimension. Besides, the depthwise convolution with a large size kernel 7×7 is positioned before the dense 1×1 convolutional layer.

5) Micro Design: Several per-layer enhancements are implemented within each block at a micro scale. These enhancements include replacing the rectified linear unit (ReLU) [42] function with the Gaussian error linear unit (GELU) [43], aligning with the activation function used in advanced transformers. Furthermore, the number of activations per block is streamlined to one. Following this adjustment, the batch normalization (BN) [44] layer is substituted with layer normalization (LN) [45], an alternative normalization choice in transformers. In addition, the number of normalization layers used per block is consolidated into a single layer.

B. Efficient Quantization Adapter Module

As a typical technique of PEFT, adapters aim to learn task-specific feature modulations on initially hidden representations by introducing extra trainable parameters into a pretrained foundation model that has been frozen.

Considering an intermediate hidden representation $\mathbf{h}$ generated by a certain layer with input in a pretraiend backbone, the adapter module learns $\Delta\mathbf{h}$ and updates $\mathbf{h}$ as

$$\mathbf{h} \leftarrow \mathbf{h} + \alpha \cdot \Delta\mathbf{h} \tag{3}$$

where α represents a gating function or a scalar.

Previous adapter methods are mainly designed for NLP tasks and transformer architectures; it is nontrivial to design effective adapter methods for ConvNet-based foundation model. In this article, we propose an EQAM to balance precision and efficiency. EQAM significantly enhances the computational and storage efficiency of PEFT techniques

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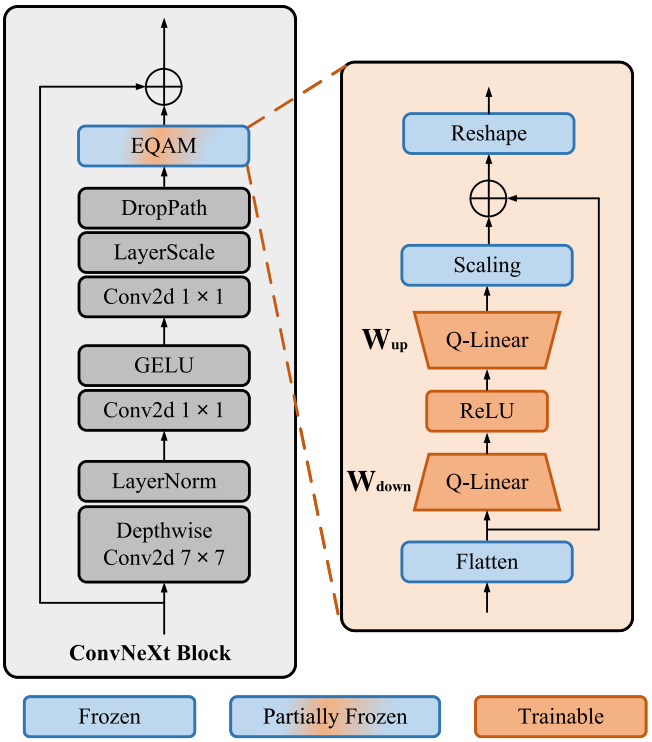


Fig. 3. Diagram illustrating the embedding of the proposed EQAM into the ConvNeXt block, with EQAM being inserted behind the DropPath regularization of each ConvNeXt block. EQAM consists of two Q-Linear layers, a ReLU nonlinear activation layer, and operations including flattening, scaling, and reshaping. During fine-tuning, only the parameters of the Q-Linear layers and ReLU are updated.

by integrating an innovative quantization strategy into the adapter module. It innovatively combines quantization and adapter optimization, achieving resource-efficient utilization while maintaining model performance.

As illustrated in Fig. 3, the proposed EQAM consists of two Q-Linear layers with a nonlinear ReLU activation function in between. In addition, it includes operations such as flattening, scaling, and reshaping. Unlike conventional linear layers applied by other adapter methods, Q-Linear is designed to pursue higher computational efficiency. We apply the clustering-based quantization to reduce the bit-width of linear layer weights. The b-bit quantization process can be conceptualized as the partitioning of the real number space $\mathbb{R}$ into $B = 2^b$ nonoverlapping sets $\{\mathcal{U}_1, \ldots, \mathcal{U}_B\}$. In this article, the bit-width b is set to 8. These sets are associated with a codebook containing B codes $\{c_1, \ldots, c_B\}$. The quantization function $\mathcal{Q}(\cdot)$ quantizes all the values in $\mathcal{U}_j$ to c_j

$$\mathcal{Q}(w) = c_j \text{ if } w \in \mathcal{U}_j. \tag{4}$$

We introduce 1-D clustering to minimize the quantization errors

$$\min_{c_1,\ldots,c_B,\mathcal{U}_1,\ldots,\mathcal{U}_B} \sum_{i=1}^{m} |w_i - \mathcal{Q}(w_i)| \tag{5}$$

where w_i denotes an element of a weight $\mathbf{W}$, and m represents the number of elements in $\mathbf{W}$.

Assuming that the parameters in $\mathbf{W}$ follow a Gaussian distribution, we standardize the weights based on the mean and variance. Subsequently, we perform quantization on each weight that has been standardized using the precomputed $\{c_1, \ldots, c_B\}$ and $\{\mathcal{U}_1, \ldots, \mathcal{U}_B\}$. Then we revert the weights to their original means and variances through de-standardization

$$w_i' = \frac{w_i - \mu}{\sigma} \tag{6}$$

$$\hat{w}_i' = \mathcal{Q}(w_i') = c_j \text{ if } w_i' \in \mathcal{U}_j \tag{7}$$

$$\hat{w}_i = \hat{w}_i{}' \cdot \sigma + \mu \tag{8}$$

where $\mu = \text{MEAN}(\{w_i\}_{i=1}^m), \sigma = \text{STD}(\{w_i\}_{i=1}^m)$.

In the entire quantization process, only the quantization operation $\mathcal{Q}$ is not differentiable. To tackle this issue, we use the straight-through estimator (STE) to approximate the gradient. For $\forall w_i, w_k \in \mathbf{W}$, the collective gradient is computed as follows:

$$\frac{\partial \hat{w}_i}{\partial w_k} = \begin{cases} 1 + \dfrac{w_i'\left(\hat{w}_i{}' - w_i'\right)}{m}, & \text{if } i = k \\ \dfrac{w_k'\left(\hat{w}_i{}' - w_i'\right)}{m}, & \text{otherwise} \end{cases}. \tag{9}$$

Formally, let the input feature map of the EQAM module be denoted as $\mathbf{F_1} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent the height, width, and channel dimensions of $\mathbf{F_1}$, respectively. The tensor $\mathbf{z}$ is initially flattened along the channel dimension to form a 2-D feature map $\hat{\mathbf{F}}_\mathbf{1}$

$$\hat{\mathbf{F}}_\mathbf{1} \in \mathbb{R}^{D\times C} = \text{Flatten}\ (\mathbf{F_1}) \tag{10}$$

where $D = H \times W$.

Then, we set the learnable weight of the two Q-Linear layers as $\mathbf{W}_{\text{down}} \in \mathbb{R}^{C\times c}$ and $\mathbf{W}_{\text{up}} \in \mathbb{R}^{c\times C}$ ($c \ll C$), aiming to adjust the space size of the low-dimensional representation through down-projection and up-projection. Ignoring the bias parameters for simplicity, the projected representation $\tilde{\mathbf{F}}_\mathbf{1}$ can be formulated as

$$\tilde{\mathbf{F}}_\mathbf{1} = \text{ReLU}\left(\hat{\mathbf{F}}_\mathbf{1}\mathbf{W}_{\text{down}}\right)\mathbf{W}_{\text{up}} \cdot \mathbf{s} + \hat{\mathbf{F}}_\mathbf{1} \tag{11}$$

where the scaling scalar $\mathbf{s}$ is set to 1 and the hidden feature dimension c is set to 8.

Finally, $\tilde{\mathbf{F}}_\mathbf{1}$ is reshaped into a 3-D feature map of the same spatial size as $\mathbf{F_1}$

$$\tilde{\mathbf{F}}_\mathbf{1}' \in \mathbb{R}^{H\times W\times C} = \text{Reshape}\ (\tilde{\mathbf{F}}_\mathbf{1}) \tag{12}$$

where the feature map generated by the EQAM module is denoted as $\tilde{\mathbf{F}}_\mathbf{1}'$.

C. Context-Aware Prompt Module

As the pioneer in effectively incorporating visual prompts into PEFT, VPT [24] introduces only a limited number of trainable parameters into the input space and overlooks task-specific contextual information critical for the interpretation of remote sensing images. In complex remote sensing scenarios, this approach demonstrates suboptimal performance as it inadequately considers contextual information and interfeature relationships. To address the aforementioned shortcomings, we propose the context-aware prompting module (CAPM), designed to generate trainable prompt parameters for contextual awareness. Subsequently, these prompts are used during the adjustment process to guide the model.

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Given the feature map $\mathbf{F_2} \in \mathbb{R}^{H\times W\times C}$ generated by a certain ConvNeXt block, we commence using a global average pooling (GAP) across spatial dimensions, resulting in the extraction of a feature embedding $\mathbf{v} \in \mathbb{R}^C$. Subsequently, $\mathbf{v}$ is processed through a linear layer responsible for channel downscaling and a softmax operation, producing a condensed prompt-weight $\varepsilon \in \mathbb{R}^L$. Overall, the above process is summarized as

$$\varepsilon = [\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_L] = \text{Softmax}(\text{Linear}(\text{GAP}(\mathbf{F}_2))). \quad (13)$$

Then prompt components $\mathbf{P} \in \mathbb{R}^{L\times \hat{H}\times \hat{W}\times \hat{C}}$ consisting of a set of learnable parameters are introduced to interact with the incoming prompt-weight ε to embed context information. To maintain a low computational cost for the CAPM, we set L to 5 and $\hat{H}, \hat{W}, \hat{C}$ to 8. In this manner, CAPM dynamically predicts weights based on contextual information from the input feature map, applying them to the prompt components to generate conditionally input-dependent prompts $\mathbf{P}_w$. In addition, CAPM establishes a shared space to facilitate the exchange and sharing of relevant knowledge among prompt components

$$\mathbf{P}_w = \sum_{l=1}^{L} \varepsilon_l \cdot \mathbf{P}_l. \quad (14)$$

Due to the mismatch in spatial dimensions between the feature map $\mathbf{F_2}$ and prompts $\mathbf{P}_w$, we use bilinear upsampling along with a $\text{Conv}_{1\times1}$ operation to upsample the prompts to the same spatial size as the input feature. Subsequently, the generated prompts from the CAPM module are added to the input feature through skip connections, facilitating the interaction between the feature and prompt information. This enables the efficient learning of task-specific representations during the fine-tuning process

$$\mathbf{F}_2' = \mathbf{F_2} + \text{Conv}_{1\times1}(\text{Upsample}(\mathbf{P}_w)) \quad (15)$$

where $\mathbf{F}_2'$ represents the feature map integrated with prompts.

IV. Experiments

In this section, we perform comprehensive experiments to assess the efficiency and effectiveness of our proposed UPetu framework in the context of remote sensing scene classification task.

A. Dataset

Three classic remote sensing scene classification datasets are used to comprehensively evaluate the interpretative ability of the proposed UPetu framework.

1) UCM [46]: The UCM dataset, designed for remote sensing studies in land-use imagery, consists of 2100 remote sensing scene images sourced exclusively from the United States Geological Survey (USGS) National Map Urban Area Imagery series. The RGB images possess a pixel resolution of 0.3 m and maintain a uniform size of 256 × 256 pixels. Derived from diverse urban areas across the United States, the dataset encompasses 21 typical land-use categories. Each category is uniformly represented by 100 images, ensuring a balanced and comprehensive dataset for in-depth analysis in the domain of remote sensing and land-use classification. Serving as a representative collection for scene classification, the dataset offers a comprehensive exploration of remote sensing challenges.

2) AID [47]: With images collected from various sensors on the Google Earth (GE) imager, the AID dataset comprises 10 000 images, each with a pixel size of 600 × 600. Covering a diverse range of 30 scene categories, the dataset presents a unique challenge for scene classification due to its high intraclass diversities. Every category within the dataset comprises an image count that varies between 220 and 400, providing a substantial and varied collection for detailed analytical pursuits in the field. The spatial resolution of the images varies from 0.5 to 8 m, providing a comprehensive representation of real-world scenarios. Notably, the AID dataset exhibits a combination of small interclass variability, large intraclass variability, and an imbalanced distribution of categories, making it a challenging benchmark for contemporary scene classification methods.

3) NWPU-RESISC45 [48]: The NWPU-RESISC45 dataset for remote sensing image scene classification, developed by Northwestern Polytechnical University, comprises 31 500 images with a pixel size of 256 × 256. It covers 45 diverse scene categories, with each category including 700 images, thereby establishing a comprehensive evaluation framework. This dataset poses challenges such as notable intraclass differences and interclass similarities, incorporating variations in translation, spatial resolution, viewpoint, object pose, illumination, background, and occlusion. The remote sensing images in NWPU-RESISC45 showcase a spatial resolution ranging from approximately 0.2 to 30 m, contributing to the dataset's enhanced complexity. This diversity makes the dataset a valuable resource for advancing research in remote sensing image classification.

B. Detailed Experimental Settings

In continuation of the research presented in [53], we select the ConvNeXt-B model pretrained on the GeoSense dataset as the remote sensing foundation model. This model exhibits enhanced capabilities in capturing distinctive features in remote sensing images, outperforming conventional pretrained models designed for natural images.

All the experiments are conducted using the PyTorch framework. The computational hardware used consisted of four NVIDIA A800 GPUs, with a batch size set to 32 for each GPU. The training duration spans 100 epochs, with an initial warm-up period conducted during the first five epochs. The optimization of the foundation model is executed using the AdamW optimizer until convergence is attained, where the learning rate and weight decay are set to 4e-4 and 0.05, respectively.

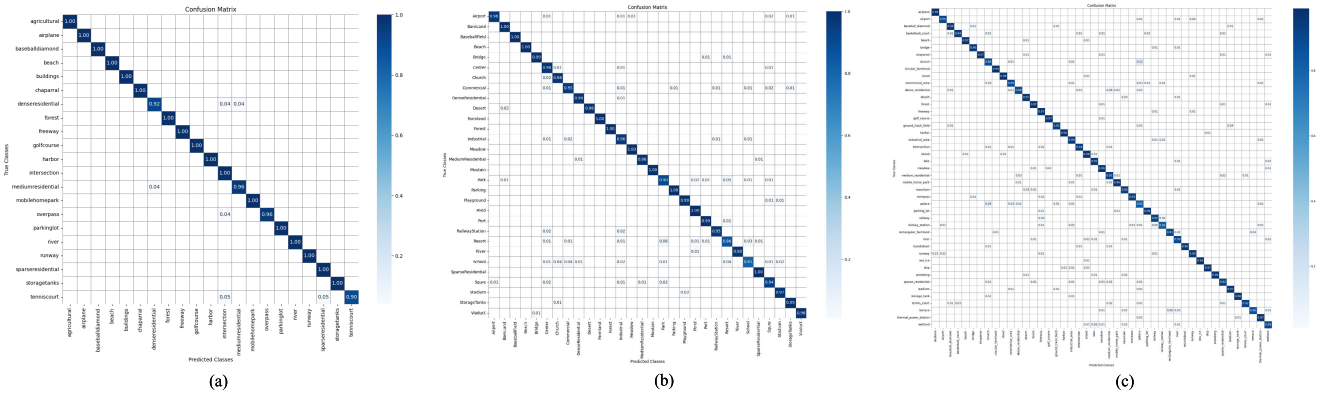
The training sample proportions in the UCM dataset are configured as TR = 50% and TR = 80%, correspondingly. For the AID dataset, the training ratios are set at TR = 20% and TR = 50%. Meanwhile, the NWPU-RESISC45 dataset uses a training rate of TR = 10% and TR = 20%. To ensure reliable and consistent results, each experiment is meticulously repeated three times. Besides, overall accuracy (OA) is used

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TABLE I

PERFORMANCE COMPARISON OF PEFT METHODS ON THREE REMOTE SENSING BENCHMARK SCENE CLASSIFICATION DATASETS. BOLD INDICATES THE BEST METRICS AMONG ALL METHODS. THE REPORTED TRAINABLE PARAMETERS REPRESENT THE AVERAGE CALCULATED OVER THE THREE DATASETS, AND THE TOTAL TRAINING TIME IS THE SUM OF THE TRAINING TIMES ACROSS ALL SETTINGS

Method	Total Training Time	Trainable Params (M)	UCM		AID		NWPU-RESISC45	
			TR=50%	TR=80%	TR=20%	TR=50%	TR=10%	TR=20%
Linear-probing	00:38:13	**0.03**	95.53	95.95	92.66	94.64	86.26	88.44
Bias-tuning	01:06:26	0.15	96.43	96.74	93.70	93.94	88.36	90.08
Full-tuning	01:29:44	87.59	**98.48**	**99.29**	**96.19**	**96.96**	**93.00**	**94.48**
Lora [49]	01:02:37	1.11	97.67	98.37	95.27	96.08	91.58	92.74
SSF [50]	**00:52:03**	0.37	97.05	97.86	94.42	95.98	89.84	91.89
Conv-Adapter [51]	00:54:19	1.22	98.08	98.27	95.59	96.76	91.65	93.38
AdapFormer [52]	01:08:22	1.26	97.72	98.10	93.96	96.22	90.23	92.51
VPT-Deep [24]	00:58:34	**0.11**	97.36	97.98	94.68	96.50	91.35	92.67
UPetu (ours)	00:53:11	0.65	**98.52**	**99.05**	**96.29**	**97.06**	**92.13**	**93.79**



(a) (b) (c)

Fig. 4. Classification confusion matrix for our proposed UPetu on (a) UCM-82, (b) AID-55, and (c) NWPU-28 datasets, respectively.

to evaluate performance, which is a widely adopted criterion in scene classification research.

C. Experimental Result Analysis

Table I presents the results of our proposed method across various datasets, comparing it with three conventional fine-tuning algorithms (i.e., linear-probing, bias-tuning, and full-tuning) and five state-of-the-art PEFT methods (i.e., Lora [49], SSF [50], Conv-Adapter [51], AdaptFormer [52], and VPT-Deep [24]).

Table I reveals that compared with the other five state-of-the-art PEFT methods, our UPetu consistently achieves superior classification performance across all the settings on the three remote sensing datasets. Furthermore, UPetu demonstrates a relatively low number of trainable parameters, specifically 0.65 M. This indicates efficient utilization of model parameters, leading to reduced computational costs during both training and inference. Particularly, under the settings of limited training samples, the proposed UPetu outperforms full-tuning (98.52% versus 98.48% on UCM-55, 96.29% versus 96.19% on AID-28, and 97.06% versus 96.96% on AID-55), underscoring its practical applicability and potential in real-world remote sensing scenarios with constrained annotations. Table I also presents a comparative analysis of training times across various PEFT methods, highlighting the efficiency of the proposed UPetu method. Notably, UPetu demonstrates a competitive total training time of 00:53:11, which is significantly shorter than full-tuning and slightly outperforms most of the state-of-the-art PEFT methods, including Lora and AdapFormer. This efficiency underscores UPetu's effectiveness, especially considering its superior performance metrics across all the datasets, which it achieves with considerably less training time than the most intensive full-tuning method. Besides, Fig. 4 displays the classification confusion matrices for the UCM-82, AID-55, and NWPU-28 datasets, illustrating the predictive outcomes of the pretrained ConvNeXt tuning with our proposed UPetu framework across diverse categories. The empirical findings substantiate the exceptional performance of our approach across nearly all the scene categories.

In addition, the t-SNE technique is used to visualize the intermediate features of fine-tuned foundation models. Through t-SNE visualization, the distribution of intermediate features in the model can be presented on a 2-D plane, facilitating a clearer observation of the data's clustering structure. As illustrated in Fig. 5, both linear-probing and bias-tuning tend to output mixed features. In contrast, the full-tuning strategy exhibits superior performance in projecting features compared with the aforementioned two methods. However, it demands a substantial amount of computational resources to fine-tune the parameters of the entire network. Furthermore, in comparison to AdaptFormer and Lora, the proposed UPetu

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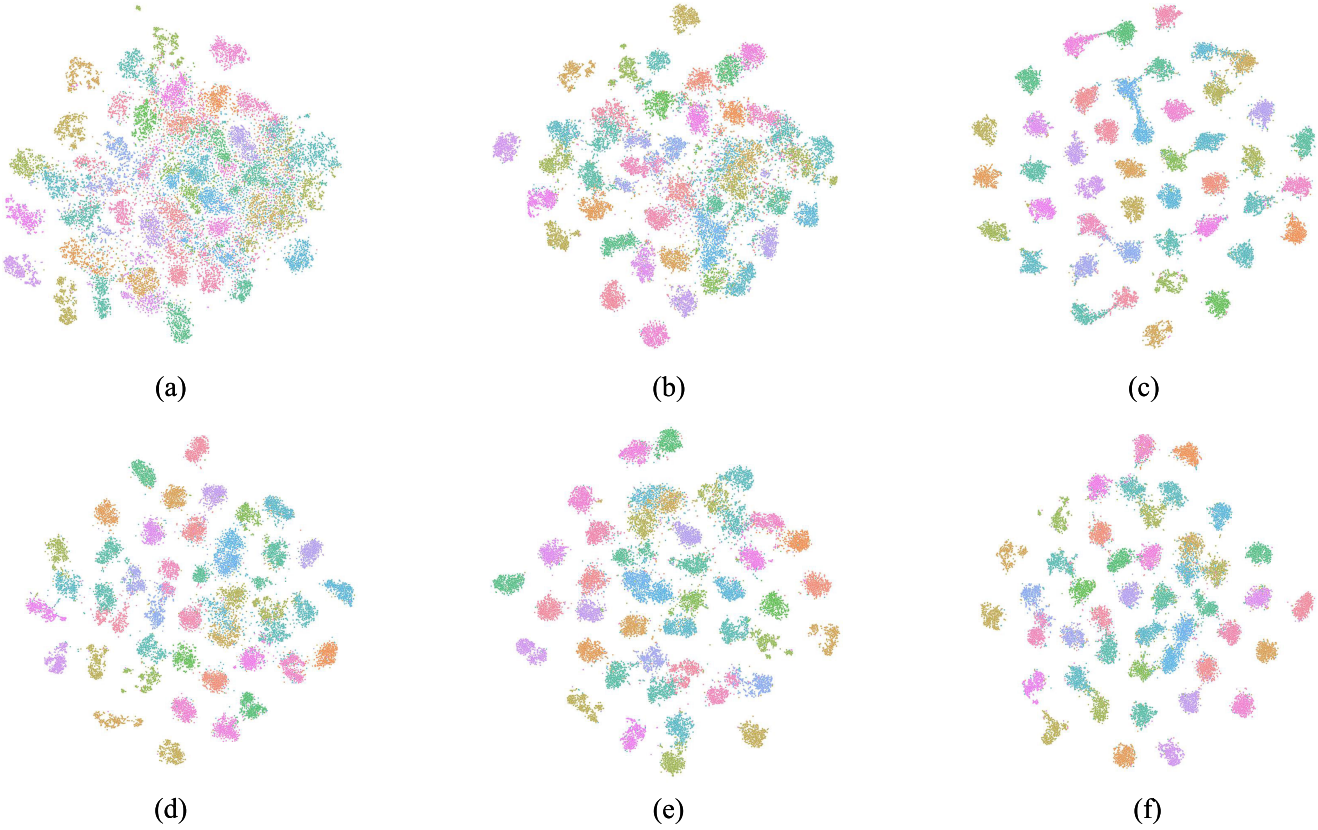


Fig. 5. t-SNE visualization of the NWPU-28 test dataset using features extracted from the last convolutional layer of the fine-tuned ConvNeXt backbone network. (a) Linear-probing. (b) Bias-tuning. (c) Full-tuning. (d) AdaptFormer. (e) Lora. (f) UPetu.

contributes to ConvNeXt generating more separable representations with fewer learnable parameters. This substantiates that UPetu aids the foundation model in effectively capturing scene information during the learning process, forming meaningful representations in the intermediate feature space.

D. Ablation Study

In this section, we demonstrate the effectiveness of the proposed UPetu and its submodules by conducting comprehensive ablation studies on the three remote sensing scene classification datasets.

1) Effectiveness of Submodules: In Table II, we showcase the scene classification performance of Upetu and its two submodules on the UCM, AID, and NWPU-RESISC45 datasets. Here, "UPetu_w_EQAM" and "UPetu_w_CAPM" indicate instances of Upetu equipped only with EQAM and CAPM modules, respectively. Table II clearly indicates that the whole UPetu has nearly achieved optimal performance across all the settings on the three datasets, highlighting the positive impact of each submodule on scene classification performance.

Furthermore, it can be observed that despite "UPetu_w_EQAM" having fewer trainable parameters, its classification performance is consistently superior across the three datasets compared with "UPetu_w_CAPM." Particularly, at TR = 50% on the AID dataset, its performance even surpasses that of full-tuning. As the proportion of training data increases, the performance of "UPetu_w_EQAM" gradually approaches that of full-tuning. In the setting with TR = 80% on the UCM dataset, "UPetu_w_CAPM" exhibits outstanding performance, achieving the same level of performance as full-tuning.

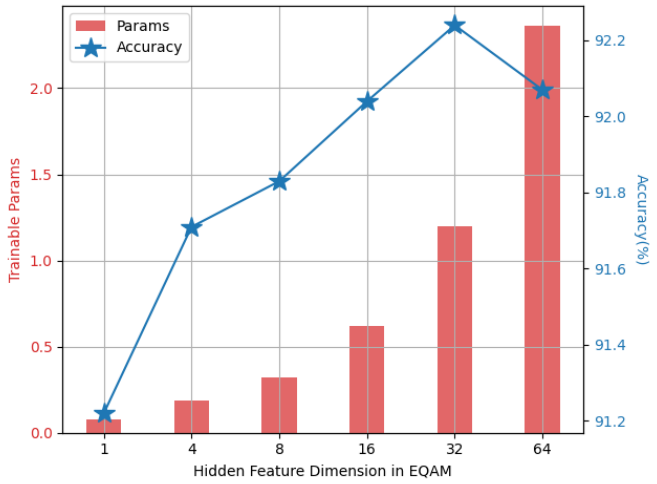


Fig. 6. Ablation study of the hidden feature dimension c in EQAM on the NWPU-19 dataset.

2) Impact of Hidden Feature Dimension in EQAM: We investigated the impact of hidden feature dimensions in EQAM on both the number of trainable parameters and classification accuracy on the NWPU-19 dataset. These dimensions fundamentally determine the latent representation space in which the foundation model learns to capture encoded patterns and

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TABLE II

EFFECTIVENESS OF THE PROPOSED UPETU FRAMEWORK AND ITS SUBMODULES EQAM AND CAPM. BOLD INDICATES THE BEST METRICS AMONG ALL METHODS. REPORTED TRAINABLE PARAMETERS REPRESENT THE AVERAGE CALCULATED ACROSS THE THREE DATASETS

Method	Trainable Params (M)	UCM		AID		NWPU-RESISC45	
		TR=50%	TR=80%	TR=20%	TR=50%	TR=10%	TR=20%
UPetu_w_EQAM	**0.32**	98.29	98.57	95.71	**97.12**	91.83	93.58
UPetu_w_CAPM	0.36	96.96	**99.05**	95.76	96.94	91.12	92.56
UPetu	0.65	**98.52**	**99.05**	**96.29**	97.06	**92.13**	**93.79**

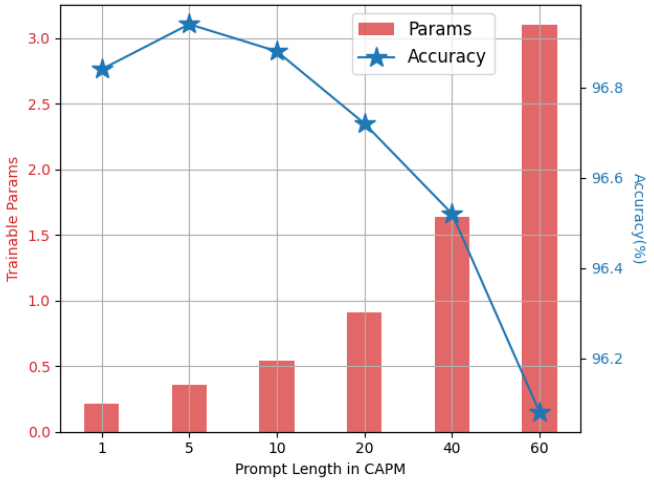


Fig. 7. Ablation study of the prompt length L in CAPM on the AID-55 dataset.

TABLE III

PERFORMANCE COMPARISON OF PEFT METHODS ON THE ISPRS POTSDAM BENCHMARK REMOTE SENSING SEMANTIC SEGMENTATION DATASET. BOLD INDICATES THE BEST METRICS AMONG ALL METHODS

Method	Trainable Params (M)	Potsdam		
		aAcc(%)	mIoU(%)	mAcc(%)
Head-tuning	**34.42**	87.97	78.61	87.66
Bias-tuing	34.56	89.73	81.30	89.31
Full-tuing	121.99	**91.76**	**84.37**	**91.02**
Lora [49]	35.49	90.55	82.48	90.00
SSF [50]	34.50	89.76	81.32	89.34
Conv-Adapter [51]	35.60	90.29	82.11	89.80
AdaptFormer [52]	35.63	88.29	79.02	88.80
VPT-Deep [24]	**34.49**	89.52	81.14	89.17
UPetu (ours)	35.04	**91.36**	**83.17**	**90.60**

features from the input data. The experimental design involved varying the hidden feature dimension across discrete values, specifically 1, 4, 8, 16, 32, and 64. The impact on the model was evaluated in terms of two critical aspects: the growth in trainable parameters and its corresponding effect on predictive accuracy.

The results, depicted in Fig. 6, reveal a noteworthy trend. With an increase in the hidden feature dimension c, there is a corresponding surge in the number of trainable parameters. Notably, trainable parameters escalate from 0.08 M at a hidden feature dimension of 1–2.36 M at a dimension of 64, suggesting a linear relationship between the hidden feature dimension and the model's parameter complexity. Simultaneously, it is observed that as the hidden feature dimension continues to increase, classification accuracy rises until $c = 32$ achieves peak performance. A marginal decrease is then noted at $c = 64$. In this study, we opted for a hidden feature dimension of 8 to strike a favorable balance between parameter efficiency and accuracy.

3) Impact of Prompt Length in CAPM: We similarly investigated the impact of prompt length on the number of trainable parameters and classification accuracy within the CAPM using the AID-55 dataset, as shown in Fig. 7. The prompt length serves as a representation of the complexity of contextual information that the introduced prompts can capture. With an extension in prompt length, there is an observed increasing trend in the number of trainable parameters, indicating an augmentation in model complexity and the potential for acquiring richer representations.

However, counterintuitively, an elongation of prompt length leads to a decrease in accuracy. Classification accuracy peaks at a prompt length L of 5 and gradually diminishes thereafter. This phenomenon suggests that as prompt length increases, the foundation model might overly emphasize noise or specific local patterns present in the training data, neglecting broader data structures. The excessive prompts cause the model to overly rely on specific patterns within the training data, limiting its ability to generalize effectively to previously unseen data.

V. DISCUSSION

In this section, we further validate the effectiveness and generalization capability of the proposed UPetu framework in the context of dense prediction tasks.

A. Semantic Segmentation

1) Dataset: The International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam[1] dataset is selected as the benchmark dataset for semantic segmentation task.

Potsdam: The Potsdam dataset published by ISPRS Commission WG *II*/4 comprises 38 high-resolution true orthophoto (TOP) tiles, each of fixed size 6000 × 6000 with a spatial resolution of 0.5 m. For our study, 24 tiles are used for training, and the remaining 14 tiles are reserved for testing. The dataset includes manual annotations for six categories: impervious surfaces, building, low vegetation, tree,

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

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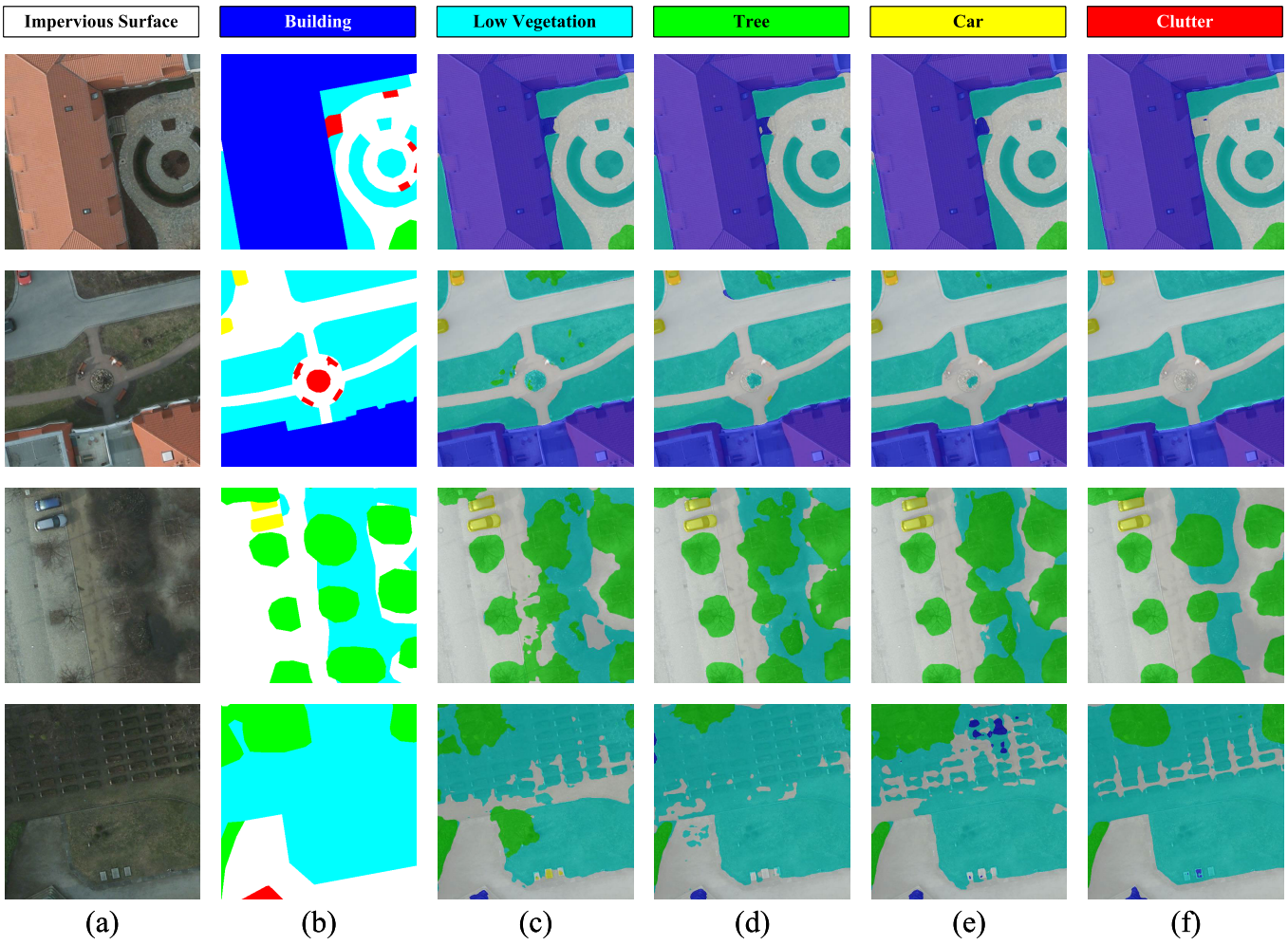


Fig. 8. Visual results of the proposed UPetu and the compared PEFT methods on the Potsdam test set. (a) Images. (b) Ground truths. (c) AdaptFormer. (d) Lora. (e) SSF. (f) UPetu.

TABLE IV

PERFORMANCE COMPARISON OF PEFT METHODS ON THE LEVIR-CD BENCHMARK REMOTE SENSING CHANGE DETECTION DATASET. BOLD INDICATES THE BEST METRICS AMONG ALL METHODS

Method	Trainable Params (M)	LEVIR-CD		
		Precision(%)	Recall(%)	F_1(%)
Head-tuning	**0.15**	89.43	80.63	84.45
Bias-tuing	0.28	90.57	81.81	85.62
Full-tuing	87.72	**94.45**	**91.87**	**93.12**
Lora [49]	1.21	91.80	83.20	86.95
SSF [50]	0.38	**91.97**	82.20	86.36
Conv-Adapter [51]	1.33	92.07	83.86	87.46
AdaptFormer [52]	1.36	91.51	83.82	87.22
VPT-Deep [24]	**0.23**	91.66	83.24	87.09
UPetu (ours)	0.77	91.91	**85.66**	**88.50**

car, and clutter. Evaluation metrics such as aAcc, mIoU, and mAcc are used, with "Clutter" excluded from the assessment.

2) Detailed Experimental Settings: All the experiments are conducted using the MMSegmentation framework and six NVIDIA A800 GPUs. The batch size is set to two per GPU. We use the AdamW optimizer for model training, setting the initial learning rate to 1e-4 and applying a weight decay of 0.05. The training incorporates a staged learning rate decay strategy, with a total of 40 000 iterations.

For the images in the Potsdam dataset, we adopt a sliding window strategy to crop them into patches of size 512 × 512, with a stride of 256 × 256 pixels. During the training phase, basic data augmentation techniques are applied, including random scaling, random flipping, and photometric distortions. In addition, UperNet [54] is selected as the semantic segmentation network.

3) Experimental Result Analysis: As shown in Table III, we evaluate the proposed UPetu over the semantic segmentation task with pretrained ConvNeXt-B. Compared with head-tuning and bias-tuning, full-tuning undeniably achieved the optimal segmentation performance; however, this came at the cost of a substantial trainable parameter size of 121.99 M. The proposed UPetu exhibits a trainable parameter size of approximately 1/4 compared with full-tuning, yet achieves segmentation performance comparable to that of full-tuning. In addition, UPetu surpasses all the compared PEFT methods in three evaluation metrics: aAcc, mIoU, and mAcc. The visualization results of Fig. 8 also demonstrate

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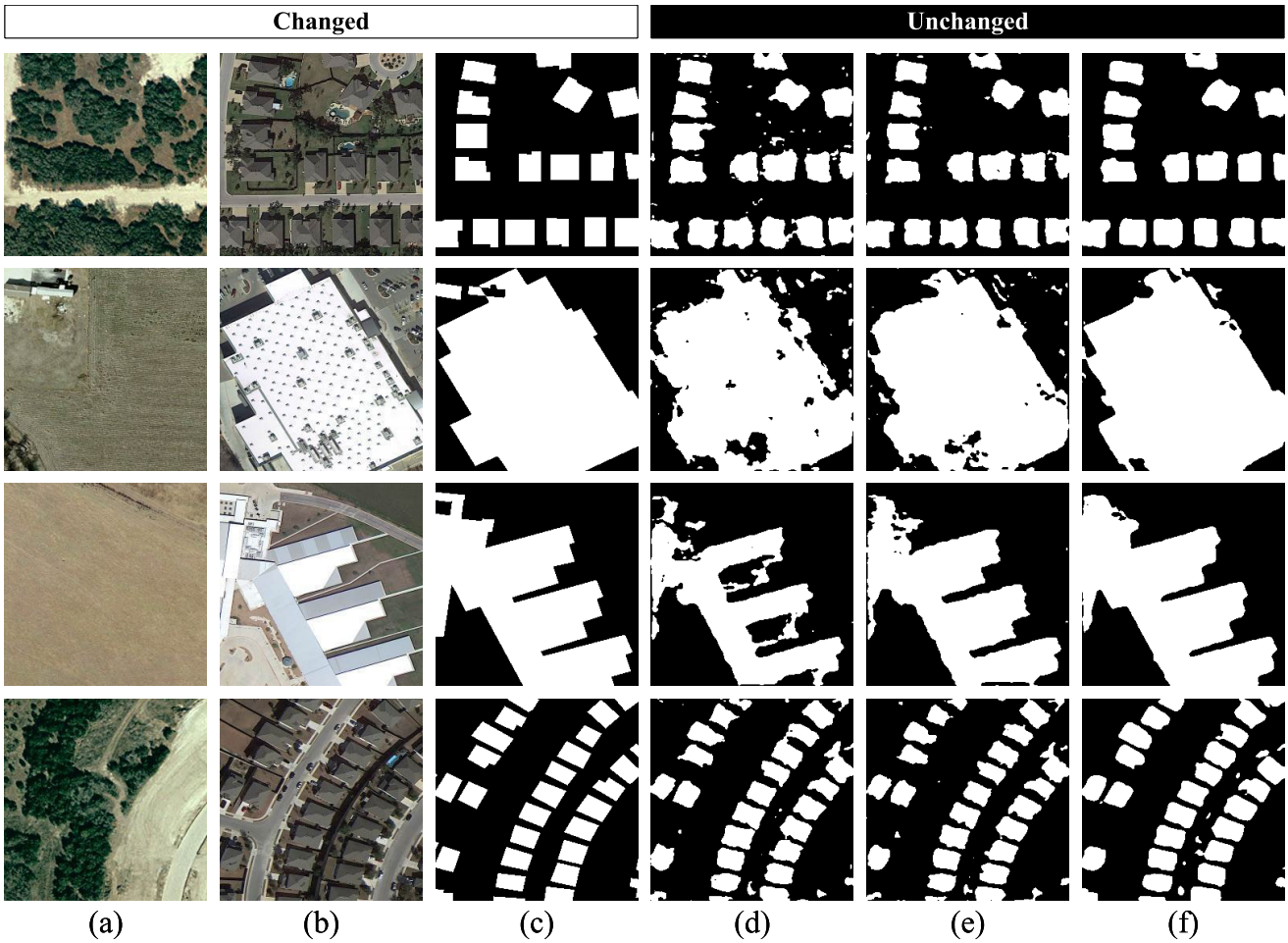


Fig. 9. Visual results of the proposed UPetu and the compared PEFT methods on the LEVIR-CD test set. (a) T0 images. (b) T1 images. (c) Ground truths. (d) AdaptFormer. (e) Conv-adapter. (f) UPetu.

that our approach achieves more accurate segmentation results compared with other methods.

B. Change Detection

1) Dataset: The standard dataset LEVIR-CD [55] is chosen for evaluating the performance of UPetu in the change detection task.

LEVIR-CD: The LEVIR-CD dataset is a newly developed, extensive-scale dataset tailored for binary change detection in remote sensing. All the images in the LEVIR-CD dataset are obtained through the GE application programming interface (API), covering the state of Texas in the United States over the period from 2002 to 2018. The dataset comprises a total of 637 image pairs, with each image having dimensions of 1024 × 1024 pixels and providing a high spatial resolution of 0.5 m. Among these pairs, 445 are designated for training, 64 for validation, and the remaining 128 for testing purposes. It is noteworthy that the primary changes observed in the dataset stem from artificial structures, encompassing a total of 31 333 distinct entities representing building alterations. The comprehensive assessment is conducted by introducing metrics precision, recall, and F_1 as evaluation metrics.

2) Detailed Experimental Settings: All the experiments are conducted using the Open-CD framework and implemented on four NVIDIA A800 GPUs. We use the AdamW optimizer with a learning rate of 0.0001 and the weight decay set to 0.05. The batch size is set to eight, and the training schedule spans 20 000 iterations, incorporating a dynamic learning rate adjustment strategy.

The images sourced from the LEVIR-CD dataset are subjected to a cropping process, resulting in nonoverlapping patches with dimensions of 256 × 256 pixels. To enhance the diversity of our dataset, we use conventional data augmentation techniques, including random rotation, cropping, and flipping. These augmentation strategies are uniformly applied to both the images within every training pair. Besides, BIT [56] is chosen as the change detection head.

3) Experimental Result Analysis: The comparative experimental results with other fine-tuning methods on the remote sensing change detection task are presented in Table IV. It can be observed that our UPetu, with only 0.77 M trainable parameters, outperforms all the compared PEFT methods in both recall and F_1 score metrics, while ranking second in precision, just behind SSF. Moreover, we present visualization results as shown in Fig. 9. Compared with AdaptFormer and Conv-Adapter, UPetu exhibits fewer misclassifications and false positives, along with smoother boundaries of predicted buildings.

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VI. Conclusion

In this article, we propose an innovative PEFT framework, UPetu, which significantly reduces the computational and storage requirements during fine-tuning of foundation models for downstream remote sensing tasks. The UPetu framework integrates two complementary key modules: the EQAM and CAPM. EQAM is specifically designed to learn task-specific feature representations during fine-tuning, effectively reducing the number of model parameters through its innovative quantization mechanism without sacrificing performance, which is particularly critical in resource-constrained remote sensing data processing. Meanwhile, CAPM endows the foundation model with the capability to dynamically generate trainable and context-dependent prompts, significantly enhancing the model's perception and processing capabilities of contextual information. Through conditionally dependent prompts, a novel way of contextual understanding is provided to the model, which is particularly important in handling complex remote sensing scenarios as it allows the model to better adapt to different tasks and environmental changes. Extensive experiments on multiple downstream tasks, including scene classification, semantic segmentation, and change detection, demonstrate the superiority of our UPetu framework in terms of efficiency and effectiveness.

While UPetu demonstrates promising results across various downstream tasks, its performance may decline when applied to remote sensing data with significant domain shift or insufficient training samples. Furthermore, UPetu's dynamic adaptability may face challenges in situations where environmental changes occur at a rate that exceeds the model's learning and adaptation capabilities. In future work, we aim to contribute toward addressing the challenge of catastrophic forgetting faced by remote sensing foundation models during incremental learning. Our focus will be on enabling remote sensing foundation models to dynamically adapt to new information, thereby minimizing the loss of proficiency in previously acquired knowledge during the incremental learning process. This endeavor involves the development of more robust and adaptive remote sensing foundation models, offering novel solutions for image interpretation in real-world scenarios.

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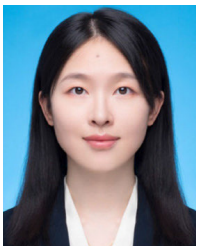
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